

Reward-Conditioned Diffusion Scenarios for Portfolio Optimization

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Abstract

This report studies whether score-based diffusion scenarios can improve portfolio construction after distributional calibration and reward-conditioned fine-tuning. The pipeline trains a variance-preserving stochastic differential equation score model on ten S&P 500 sector ETF daily log returns from 2014 to 2020, evaluates hyperparameters only on 2021 validation returns, and reserves 2022 to 2024 for final testing. Gaussian Bures-Wasserstein optimal transport is used to match generated first and second moments to training moments. A KL-penalized reverse-diffusion control network is then trained with a decision-aligned validation reward. The production evaluation passes the leakage audit with zero failures and zero warnings. The strongest fixed-scenario diffusion baseline obtains Sharpe 0.454, while the validation-selected fine-tuned models underperform out of sample in fixed mode. In rolling mode, all diffusion variants collapse toward rolling Markowitz performance, with Sharpe ratios around 0.51. The results show that the method is reproducible and mathematically coherent, but the current fine-tuning reward overfits validation-period structure and does not yet provide robust test-period alpha.

1 Introduction

Portfolio optimization depends on an estimate of the return distribution. The classical Markowitz problem estimates means and covariances from historical returns, then solves a mean-variance allocation problem (Markowitz, 1952). This is simple and transparent, but the estimated mean vector is noisy and the resulting portfolio can become highly concentrated. Distributionally robust portfolio optimization partially addresses this problem by optimizing against a neighborhood of plausible distributions rather than a single empirical estimate (Blanchet et al., 2022).

Generative models offer a different route. Instead of estimating only moments, they can produce synthetic return scenarios and let the downstream optimizer solve a sample-average problem. Score-based diffusion models are a natural candidate because they define a continuous-time noising process and learn the reverse score needed for generation (Song et al., 2021). Denoising diffusion models have been successful in high-dimensional generative modeling (Ho et al., 2020), and the SDE formulation gives a direct link to stochastic control.

Financial returns, however, are not just generic multivariate data. They exhibit stylized facts such as heavy tails, weak linear autocorrelation, volatility clustering, and unstable dependence (Cont, 2001).

A return generator can look reasonable under some marginal plots while still producing moments that imply poor portfolios. The central question in this project is therefore practical: can a diffusion model be made useful for portfolio decisions while preserving enough distributional realism to avoid pure reward hacking?

The project evaluates three connected ideas. First, train a base variance-preserving SDE model on historical sector ETF returns. Second, calibrate generated scenarios with Gaussian optimal transport so that first and second moments match the training distribution. Third, fine-tune the reverse diffusion drift with a KL-penalized portfolio reward evaluated only on validation data. The KL penalty is meant to limit how far the controlled reverse process can move from the base model.

The final results are materially different from the preliminary positive result. The production experiment evaluates both a base score model and an OT-augmented score model, and selects η separately for each family. Validation Sharpe is very high, above 3.1 for the best candidates, but the selected fine-tuned fixed-scenario strategies lose money during the test period. That mismatch is the main empirical finding.

2 Data and Experimental Protocol

2.1 Assets and Splits

The dataset contains daily log returns for ten S&P 500 sector ETFs: XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLB, and XLRE. The assets cover technology, financials, energy, health care, industrials, consumer discretionary, consumer staples, utilities, materials, and real estate. This universe is small enough for controlled experimentation but broad enough to produce nontrivial covariance and allocation structure.

The experiment uses a strict chronological split:

Split	Date range	Trading days	Use
Train	2014-01-01 to 2020-12-31	1,317	score model, scaler, OT target
Validation	2021-01-01 to 2021-12-31	252	η selection, reward evaluation
Test	2022-01-01 to 2024-12-31	756	final performance evaluation

The standardization scaler is fit only on training data. The experiment records the training range, validation range, test-use status, and scaler provenance for each model artifact. The final leakage audit reports a passing status, with zero failures and zero warnings.

The chronological split is important because the test window contains a market environment that differs sharply from the training window. The training period includes the long post-crisis expansion, the 2018 volatility episode, and the COVID-19 shock. The validation year, 2021, is a strong recovery year for many risk assets. The test period includes the 2022 inflation and rate-hiking regime, the 2023 rebound, and the 2024 continuation of concentrated equity strength. This shift creates a difficult but useful evaluation setting. A model that optimizes too directly for the validation year can select a portfolio that looks excellent in 2021 but fails in the higher-rate regime that follows.

The sector ETF universe also makes the estimation problem concrete. The assets are liquid, diversified within sector, and highly correlated during stress periods. Their return distributions are not independent, and the covariance matrix has economically meaningful structure. Technology and consumer discretionary often carry high growth exposure; utilities and consumer staples often provide more defensive exposure; energy has distinct commodity sensitivity. The optimizer must therefore decide not only between high and low volatility assets, but also between correlated factor exposures.

2.2 Backtest Design

The project compares equal-weight, rolling Markowitz, Wasserstein-robust Markowitz, base diffusion scenarios, Gaussian-OT diffusion scenarios, and fine-tuned diffusion scenarios. All reported test metrics use 2022 to 2024 data only. The final comparison stage loads the frozen eta selection before test evaluation and does not use test data for model selection.

Two backtest modes are reported. In fixed-scenario mode, each diffusion strategy uses one scenario set throughout the test period. This isolates the scenario generator as a static model of the training distribution. In rolling mode, each monthly rebalance recalibrates scenarios to the trailing 252-day mean and covariance using Gaussian OT. That mode tests whether diffusion scenarios add anything after current rolling moments are imposed.

The baseline set is designed to separate three effects. Equal weight measures the benefit of simple diversification without estimation. Rolling Markowitz measures the value and instability of plug-in moment estimation. Wasserstein-robust Markowitz measures whether shrinking toward a more conservative covariance estimate improves realized risk. The diffusion strategies measure whether scenario generation, moment calibration, and reward fine-tuning add incremental value beyond those classical baselines. Because all strategies are evaluated on the same daily return path, the comparison emphasizes allocation decisions rather than differences in data handling.

3 Diffusion Model

3.1 Forward SDE

Let $X_0 \in \mathbb{R}^d$ denote a vector of standardized daily returns, with $d = 10$. The forward variance-preserving SDE is

$$dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)} dW_t, \quad \beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min}), \quad (1)$$

with $\beta_{\min} = 0.1$ and $\beta_{\max} = 20.0$. The conditional marginal is

$$X_t|X_0 \sim \mathcal{N}(\alpha_t X_0, \sigma_t^2 I), \quad \alpha_t = \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right), \quad \sigma_t^2 = 1 - \alpha_t^2. \quad (2)$$

The score network $s_\theta(x, t)$ estimates $\nabla_x \log p_t(x)$. Training minimizes denoising score matching:

$$\mathcal{L}_{DSM}(\theta) = \mathbb{E}_{t, X_0, \epsilon} \left[\|s_\theta(\alpha_t X_0 + \sigma_t \epsilon, t) + \epsilon / \sigma_t\|_2^2 \right], \quad \epsilon \sim \mathcal{N}(0, I). \quad (3)$$

The network is a multilayer perceptron with hidden size 256, SiLU activations, and a sinusoidal time embedding of dimension 64. The production model trains for 2,000 epochs with Adam at learning rate 10^{-3} (Kingma and Ba, 2015). The final training loss is 0.1574.

3.2 Reverse Sampling

Scenarios are generated by Euler-Maruyama discretization of the reverse process. The production run uses 500 reverse steps and generates 10,000 base scenarios. The generated samples are transformed back to return units using the training-fitted scaler. The base scenario diagnostics record a Bures-Wasserstein distance of 0.0309 from the training distribution and an implied long-only MVO HHI near 1.0, meaning the raw generated moments induce an almost fully concentrated portfolio.

This concentration problem is central. A generator can have acceptable correlation plots while still producing a slightly biased mean vector. In mean-variance optimization, small mean errors can dominate the allocation because the inverse covariance amplifies the direction of perceived excess

return. The raw diffusion model underestimates volatility across assets and produces a portfolio that is effectively a single-asset bet. That is not only undesirable from a risk-management perspective; it is also evidence that distributional diagnostics must be tied to downstream decisions. A scenario model should be judged both by statistical realism and by the stability of the portfolio it implies.

3.3 Diagnostics Beyond Loss

Training loss alone is not sufficient for this application. The project evaluates generated scenarios along three dimensions. The first is marginal realism: per-asset means, standard deviations, kurtosis, and quantile plots should be close to the training returns. The second is cross-sectional realism: generated correlations and covariances should preserve the dependence structure among sectors. The third is decision realism: portfolios implied by generated scenarios should not be pathologically concentrated or unstable.

These diagnostics reveal a clear pattern. The score model learns a reasonable cross-sectional shape but compresses marginal volatility and fat tails. This is common when a simple model is trained on limited financial data with a squared denoising objective. The loss rewards average denoising accuracy, while the tails are rare and carry less direct weight. For portfolio construction, this matters because underestimated volatility can make the optimizer too aggressive. Calibration is therefore not a cosmetic postprocessing step; it is required to make the generated scenarios usable in a mean-variance allocation.

4 Optimal Transport Calibration

Gaussian OT is used as a moment-correcting postprocessor. Let generated samples have mean and covariance (μ_g, Σ_g) , and let training returns have (μ_t, Σ_t) . Under a Gaussian approximation, the Bures-Wasserstein map from generated to target moments is affine:

$$T(x) = \mu_t + A(x - \mu_g), \quad (4)$$

where

$$A = \Sigma_g^{-1/2} \left(\Sigma_g^{1/2} \Sigma_t \Sigma_g^{1/2} \right)^{1/2} \Sigma_g^{-1/2}. \quad (5)$$

This construction follows the closed-form geometry of Gaussian optimal transport and the Bures-Wasserstein metric (Bures, 1969; Bhatia et al., 2019; Peyre and Cuturi, 2019). It matches the first two moments exactly up to numerical error. It does not correct tail behavior, skewness, volatility clustering, or time dependence.

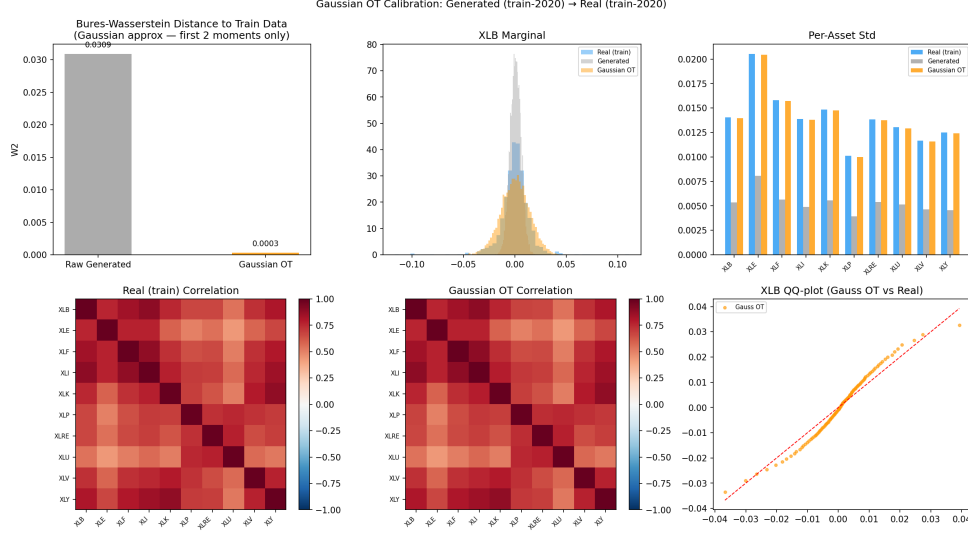
This limitation is not a weakness of the implementation. It is a property of the chosen map. A Gaussian OT map treats the source and target distributions as if their relevant structure were fully summarized by mean and covariance. For a mean-variance optimizer, that is a reasonable first-order correction because the optimizer itself uses only those two moments. For a tail-risk optimizer, it would be insufficient. In that setting, a calibration method would need to preserve or reshape distributional features such as left-tail dependence and asymmetric drawdowns.

The benefit of Gaussian OT is interpretability. If the calibrated scenarios perform differently from the raw scenarios, the source of the difference is clear: the first two moments have changed. This makes the fixed-scenario comparison easier to diagnose. Raw diffusion performance reflects the generator's uncontrolled moment bias. Gaussian-OT performance reflects a corrected mean-covariance input to the same portfolio optimizer. Fine-tuned performance then reflects a reward-conditioned change in scenario distribution.

The diagnostics show that Gaussian OT reduces the training-distance diagnostic from 0.0309 to 0.00034 and reduces covariance Frobenius error from 0.00141 to 1.75×10^{-5} . The raw generated marginal standard deviations are too low for every asset. For example, XLK has generated standard

Table 1: Scenario calibration diagnostics from the production evaluation.

Scenario set	W2 diagnostic	Covariance error	Implied HHI
Base generated	0.0309	0.00141	1.000
Gaussian OT	0.00034	0.000017	0.264
Sinkhorn subset	diagnostic only	diagnostic only	not used

**Figure 1:** Gaussian OT diagnostics. The calibration aligns first and second moments to the training distribution, but it should not be interpreted as a full correction of non-Gaussian financial structure.

deviation 0.00558 versus training standard deviation 0.01486. After Gaussian OT, the XLK standard deviation becomes 0.01476.

5 KL-Penalized Fine-Tuning

5.1 Controlled Reverse Process

Fine-tuning adds a learnable drift perturbation $\mathbf{u}_\phi(y, t)$ to the reverse SDE:

$$dY_t = \left[-\frac{1}{2}\beta(t)Y_t + \beta(t)s_\theta(Y_t, t) + \mathbf{u}_\phi(Y_t, t) \right] dt + \sqrt{\beta(t)} dW_t. \quad (6)$$

The control network is a two-hidden-layer MLP with hidden size 128. Its output is bounded as $\tanh(\text{MLP}(y, t)) \cdot 1.5$ to avoid extreme drifts.

By Girsanov's theorem, the path-space KL divergence between the controlled and base reverse processes has a quadratic control-cost form (Oksendal, 2003):

$$\text{KL}(P_\phi \| P_\theta) = \mathbb{E}_{P_\phi} \left[\int_0^T \frac{\|\mathbf{u}_\phi(Y_t, t)\|_2^2}{2\beta(t)} dt \right]. \quad (7)$$

The fine-tuning objective is

$$\max_{\phi} \mathbb{E}_{Y_0 \sim P_\phi} [R(Y_0)] - \eta \text{KL}(P_\phi \| P_\theta). \quad (8)$$

Small η permits larger drift changes. Large η keeps the model closer to the base generator.

5.2 Portfolio Validation Reward

For generated scenario batch Y_0 , the method computes batch moments $(\hat{\mu}, \hat{\Sigma})$ and a differentiable soft-Markowitz portfolio:

$$\hat{\pi} = \text{softmax} \left((\hat{\Sigma} + 10^{-4}I)^{-1} \hat{\mu} \right). \quad (9)$$

The softmax gives positive weights summing to one, but it is not the exact solution to the constrained Markowitz problem. Exact differentiable convex optimization layers would be a better future replacement (Agrawal et al., 2019).

The softmax approximation has two practical consequences. First, it guarantees long-only weights, which avoids short-sale leverage and keeps the allocation comparable to the baseline constraints. Second, it can exaggerate small differences in the unconstrained Markowitz direction. If one asset has a slightly larger estimated risk-adjusted return, the softmax can push most of the portfolio into that asset, especially when the covariance estimate is ill conditioned. The concentration penalty is included to counteract this effect, but the fixed-test results show that the penalty is not strong enough for the most aggressive selected model.

The reward applies $\hat{\pi}$ to fixed 2021 validation returns and combines annualized mean, annualized variance, empirical CVaR, and concentration:

$$R(Y_0) = 252\bar{r}_v - \frac{\lambda}{2} 252 \text{Var}(r_v) - \gamma \text{CVaR}_{95}(r_v) - \tau \sum_i \hat{\pi}_i^2, \quad (10)$$

where $\lambda = 5$, $\gamma = 0.5$, and $\tau = 0.1$. Gradients flow through $\hat{\pi}$ to the generated scenario moments and back through the controlled reverse process. Test returns are never used in this stage.

The reward is deliberately decision-aligned: generated scenarios are useful only if the portfolio they imply performs well on validation returns. This is more direct than matching stylized facts alone. It is also riskier. Validation returns are a single historical realization, not a population objective. If 2021 rewards a concentrated growth-oriented exposure, the fine-tuning step can learn to generate scenarios that justify that exposure, even if the resulting portfolio does not survive the next regime. The KL penalty limits the size of the drift perturbation, but it does not eliminate validation overfitting.

6 Eta Selection

The eta sweep evaluates $\eta \in \{0.05, 0.1, 0.5, 1.0, 5.0\}$ for both the base score model and the OT-augmented score model. Selection is based only on 2021 validation Sharpe. The selection protocol explicitly records that test metrics are not available at selection time.

The selected models are OT-augmented eta 1.0 and base eta 0.05. Both selection scores are high, but both turn out to be poor fixed-scenario test portfolios. This gap is evidence that the reward and eta selection protocol can overfit a single validation year.

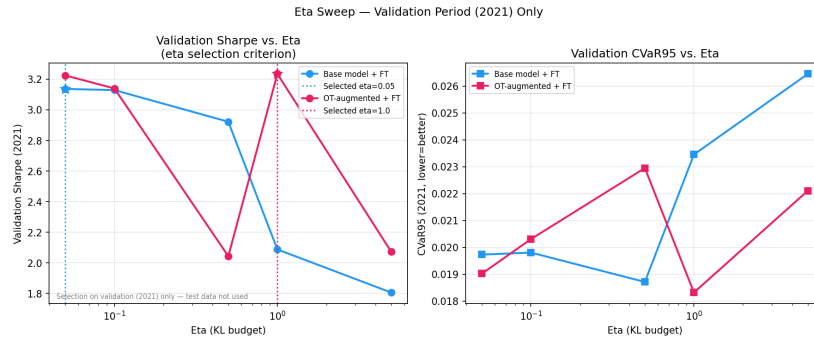
The eta table also shows that validation Sharpe alone is not a sufficient model selection criterion. The base eta 0.05 and eta 0.10 models both reach validation Sharpe above 3.1, but their HHI values are essentially one. A selection rule that jointly constrained validation Sharpe, HHI, and sensitivity to small changes in the validation window would likely reject those models. The OT-augmented eta 1.0 model is less concentrated, but its fixed-test return is still negative. This suggests that concentration is not the only failure mode. The validation reward can also select a sector tilt that is diversified but regime-specific.

7 Fixed-Scenario Results

Table 3 reports the fixed-scenario test results. Equal weight earns 5.59% annually with Sharpe 0.363. Rolling Markowitz earns 9.03% with Sharpe 0.515, but it is concentrated with HHI 0.541 and high

Table 2: Validation eta sweep. The selected eta is the highest validation Sharpe within each model family.

Model	Eta	Val. Sharpe	Ann. return	Ann. vol.	HHI
base	0.05	3.136	46.04%	14.68%	0.982
base	0.10	3.128	46.10%	14.74%	1.000
base	0.50	2.921	41.49%	14.20%	0.447
base	1.00	2.086	36.49%	17.49%	0.732
base	5.00	1.805	34.74%	19.25%	1.000
OT-aug.	0.05	3.224	44.08%	13.67%	0.714
OT-aug.	0.10	3.139	48.74%	15.53%	0.532
OT-aug.	0.50	2.042	34.97%	17.12%	0.684
OT-aug.	1.00	3.234	41.65%	12.88%	0.520
OT-aug.	5.00	2.071	34.16%	16.49%	0.587

**Figure 2:** Validation frontier used for eta selection. Test data are not used in this figure or in the eta-selection procedure.

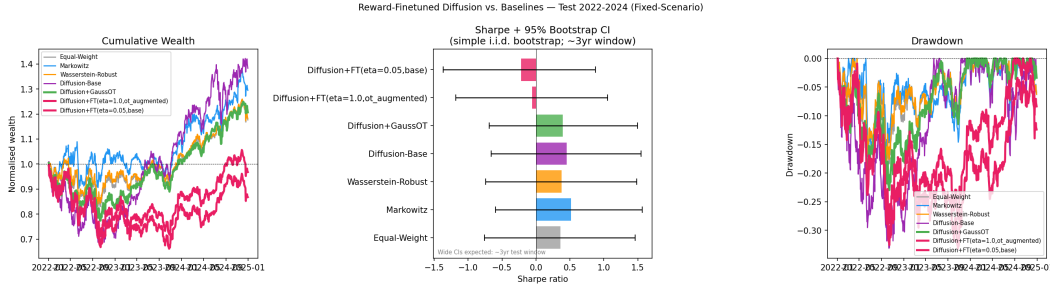
turnover. The Wasserstein-robust baseline is diversified and low turnover, but its Sharpe is only 0.380. Among diffusion strategies, the raw base scenarios produce annual return 11.48% and Sharpe 0.454, but with annual volatility 25.31%, maximum drawdown 33.13%, and HHI 1.0. This is not a usable allocation profile. Gaussian OT lowers volatility and concentration, but also lowers return, producing Sharpe 0.393. The validation-selected fine-tuned models perform worst in fixed mode: OT-augmented eta 1.0 has Sharpe -0.059, and base eta 0.05 has Sharpe -0.221.

The confidence intervals are wide. That is expected with about 756 daily test observations, especially when returns are non-Gaussian and serial dependence is possible. The simple bootstrap is useful as a warning against overclaiming, but a stationary or block bootstrap would be more appropriate for dependent financial returns (Politis and Romano, 1994). No pairwise Sharpe difference should be treated as statistically decisive.

The fixed-scenario results are still informative. They show that the base diffusion model contains enough signal to generate a high-return allocation, but not enough calibration to control risk. They also show that first-two-moment calibration improves risk characteristics but does not create superior realized performance. Most importantly, they show that validation reward optimization can move the model in the wrong direction. This is the central empirical warning: decision-aligned fine-tuning is powerful enough to change the portfolio, but the current validation design is not strong enough to guarantee that the change is robust.

Table 3: Fixed-scenario backtest, 2022 to 2024. Sharpe confidence intervals use 2,000 i.i.d. bootstrap draws.

Strategy	Ann. ret.	Ann. vol.	Sharpe	95% CI	Max DD	HHI
Equal-Weight	5.59%	15.41%	0.363	[−0.76, 1.46]	−18.41%	0.100
Markowitz	9.03%	17.54%	0.515	[−0.60, 1.57]	−17.09%	0.541
Wasserstein-Robust	5.75%	15.12%	0.380	[−0.74, 1.48]	−17.92%	0.103
Diffusion-Base	11.48%	25.31%	0.454	[−0.66, 1.55]	−33.13%	1.000
Diffusion+GaussOT	6.50%	16.56%	0.393	[−0.69, 1.50]	−22.97%	0.264
Diffusion+FT, OT-aug. eta 1.0	−1.08%	18.35%	−0.059	[−1.18, 1.06]	−28.82%	0.520
Diffusion+FT, base eta 0.05	−4.63%	20.93%	−0.221	[−1.37, 0.88]	−33.20%	0.982

**Figure 3:** Fixed-scenario test comparison. The fine-tuned strategies selected by validation Sharpe do not generalize in fixed-scenario mode.

8 Rolling Results

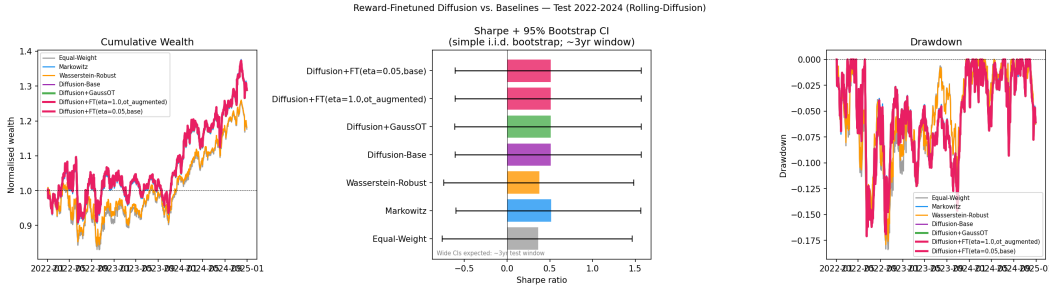
Table 4 reports the rolling-diffusion test results. Once generated scenarios are recalibrated to the trailing 252-day historical moments, all diffusion variants produce nearly the same performance as rolling Markowitz. Diffusion-Base obtains Sharpe 0.5105, Diffusion+GaussOT obtains 0.5096, the OT-augmented fine-tuned model obtains 0.5101, and the base fine-tuned model obtains 0.5106. These values are economically indistinguishable from rolling Markowitz at 0.5151.

This collapse has a clear explanation. Mean-variance optimization depends only on the estimated mean and covariance. Rolling Gaussian OT forces the scenario set to match the same trailing moments that rolling Markowitz already uses. Unless the portfolio objective uses higher-order scenario information, or the generator contributes predictive structure beyond trailing moments, the rolling diffusion strategy has little room to differ.

This result should not be read as a failure of diffusion modeling in general. It is a consequence of pairing a moment-matched scenario generator with a mean-variance optimizer. Once the rolling calibration step imposes the current mean and covariance, the remaining higher-order differences in the scenario set are mostly invisible to the optimizer. A different downstream objective could use those differences. For example, a CVaR optimizer would depend on the lower tail of the generated distribution. A drawdown-aware objective would depend on path structure. A regime-conditioned optimizer could use the generator to forecast state-dependent covariance rather than simply reproduce trailing moments.

Table 4: Rolling-diffusion backtest, 2022 to 2024. Scenarios are recalibrated to trailing 252-day moments at each rebalance using only past data.

Strategy	Ann. ret.	Ann. vol.	Sharpe	Max DD	Turnover
Equal-Weight	5.59%	15.41%	0.363	-18.41%	0.000
Markowitz	9.03%	17.54%	0.515	-17.09%	0.617
Wasserstein-Robust	5.75%	15.12%	0.380	-17.92%	0.040
Diffusion-Base	8.90%	17.44%	0.511	-17.10%	0.588
Diffusion+GaussOT	8.87%	17.40%	0.510	-17.09%	0.586
Diffusion+FT, OT-aug. eta 1.0	8.88%	17.41%	0.510	-17.09%	0.585
Diffusion+FT, base eta 0.05	8.90%	17.43%	0.511	-17.10%	0.588

**Figure 4:** Rolling-diffusion test comparison. Moment recalibration makes diffusion strategies behave almost like rolling Markowitz.

9 Interpretation

9.1 What Worked

The pipeline is reproducible and properly separated by time. The audit passed. The score model generated finite scenarios with recorded provenance. Gaussian OT did exactly what it was supposed to do: it aligned first and second moments and reduced concentration relative to raw generated scenarios. The rolling test also behaved consistently with the mathematics of the pipeline.

The project also demonstrates a useful engineering structure. Every stage is automated, including data preparation, leakage audit, score training, sampling, OT calibration, eta sweep, comparison, and report asset generation. This makes the experiment repeatable without changing the modeling protocol between runs.

9.2 What Did Not Work

The fine-tuning reward did not generalize. Validation Sharpe selected highly aggressive portfolios. The base eta 0.05 model has validation Sharpe 3.136 and HHI 0.982, then loses 4.63% annually in fixed test with HHI 0.982. That is a classic sign of validation-period overfitting and excessive concentration. The OT-augmented eta 1.0 model is less concentrated, but it still has negative fixed-test return.

The base generator also underestimates marginal volatility and fat tails before OT calibration. The generated kurtosis values in the diagnostics are near zero, while training-return kurtosis is high across assets. Gaussian OT fixes volatility but cannot fix this tail mismatch. As a result, the scenario distribution is moment-aligned but not fully realistic.

9.3 Why the Earlier Positive Claim Changed

The prior draft stated that fine-tuning achieved Sharpe 0.70 in fixed mode. The final metrics do not support that statement. The updated selection protocol evaluates both base and OT-augmented model families and chooses different etas. Under those frozen selections, fixed-test fine-tuning underperforms. The report therefore treats the final production evaluation as authoritative.

10 Limitations

The first limitation is model expressiveness. A plain MLP score network on one-day return vectors does not model temporal dynamics directly. It sees the cross-section of daily returns, not a sequence of market states. That makes it difficult to capture volatility clustering and regime dependence, both of which matter in financial data.

The second limitation is the reward. The soft-Markowitz reward is differentiable, but it can produce highly concentrated portfolios when the generated moment estimates are noisy. It optimizes validation-year performance rather than a more stable distributional objective. Stronger concentration controls, turnover-aware rewards, and nested cross-validation across multiple validation regimes would make eta selection more reliable.

The third limitation is the calibration layer. Gaussian OT is useful and transparent, but it reduces scenario calibration to moment matching. If the downstream optimizer is mean-variance, then moment matching may be enough. If the project goal is to exploit higher-order scenario structure, the optimizer must use tail or path-dependent objectives, and the generator must be evaluated on those properties.

The fourth limitation is statistical power. A three-year test window is small for Sharpe-ratio inference. The confidence intervals in Tables 3 and 4 are wide even under a simple i.i.d. bootstrap. A longer history, walk-forward validation, and block bootstrap inference are needed before making strong performance claims.

The fifth limitation is the objective mismatch between training diagnostics and final use. The score model is trained to estimate a denoising score, the OT map is evaluated by moment alignment, the fine-tuning step is evaluated by validation Sharpe, and the final report evaluates realized test performance. Each stage is reasonable in isolation, but the objectives are not identical. Future work should reduce this mismatch by defining a single decision-aware evaluation protocol that is used consistently for model selection and final assessment.

11 Future Work

The most important next step is to replace single-year eta selection with a walk-forward validation design. For example, train on 2014 to 2018, validate on 2019, train on 2014 to 2019, validate on 2020, then reserve 2021 for final model selection before testing on 2022 to 2024. This would reduce dependence on one unusually favorable validation year.

The second improvement is to use a differentiable convex optimization layer for the portfolio decision. That would align the training reward with the exact long-only constrained optimizer used in evaluation, rather than relying on a softmax approximation.

The third improvement is to condition the diffusion model on market state. A regime-conditioned score model could generate scenarios conditional on realized volatility, trend, rates, or macro indicators. That would give the rolling strategy a way to differ from simple trailing-moment Markowitz.

The fourth improvement is to evaluate objectives that actually use scenario tails. If the downstream portfolio remains mean-variance, the generator is judged mostly through means and covariances. CVaR optimization, drawdown-aware objectives, or utility functions with downside asymmetry would

better test whether diffusion scenarios add value beyond moment estimation.

12 Conclusion

The latest production run supports a sober conclusion. The project successfully builds a controlled, reproducible pipeline for diffusion-based portfolio scenarios with strict data separation, Gaussian OT calibration, and KL-penalized reward fine-tuning. The mathematical method is sound and the calibration diagnostics behave as expected. The leakage audit passes cleanly.

The empirical result is mixed. Base diffusion scenarios have high fixed-test return but unacceptable concentration and drawdown. Gaussian OT improves moment alignment and concentration, but it does not beat rolling Markowitz. The validation-selected fine-tuned models do not generalize in fixed-scenario mode. In rolling mode, Gaussian moment recalibration causes diffusion strategies to converge to rolling Markowitz performance.

The main lesson is that reward-conditioned diffusion for portfolios is promising as a framework, but the current implementation is not yet a robust trading method. The next version should focus on stronger validation, exact differentiable portfolio optimization, regime conditioning, and tail-aware objectives.

References

References

- Agrawal, A., Amos, B., Barratt, S., Boyd, S., Diamond, S., and Kolter, J. Z. (2019). Differentiable convex optimization layers. *Advances in Neural Information Processing Systems*, 32. <https://papers.neurips.cc/paper/9152-differentiable-convex-optimization-layers>.
- Anderson, B. D. O. (1982). Reverse-time diffusion equation models. *Stochastic Processes and Their Applications*, 12(3):313–326.
- Bhatia, R., Jain, T., and Lim, Y. (2019). On the Bures-Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191. <https://arxiv.org/abs/1712.01504>.
- Blanchet, J., Chen, L., and Zhou, X. Y. (2022). Distributionally robust mean-variance portfolio selection with Wasserstein distances. *Management Science*, 68(9):6382–6410. <https://pubsonline.informs.org/doi/abs/10.1287/mnsc.2021.4155>.
- Bures, D. (1969). An extension of Kakutani’s theorem on infinite product measures to the tensor product of semifinite W^* -algebras. *Transactions of the American Mathematical Society*, 135:199–212.
- Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236. <https://www.tandfonline.com/doi/abs/10.1080/713665670>.
- Ho, J., Jain, A., and Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851. <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>.
- Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations*.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.
- Oksendal, B. (2003). *Stochastic Differential Equations: An Introduction with Applications*. Springer, 6th edition.
- Peyre, G. and Cuturi, M. (2019). Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607. <https://arxiv.org/abs/1803.00567>.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313. <https://www.tandfonline.com/doi/abs/10.1080/01621459.1994.10476870>.
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. (2021). Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations*. <https://research.google/pubs/score-based-generative-modeling-through-stochastic-differential-equations/>.